

ACCOUNT CREATION / ONBOARDING – FINAL TAXONOMY v1

Following the structure established in the Identity/KYC document, I have normalized the validated candidates into **3 distinct attack families**. Variants are grouped under their underlying mechanism rather than treated as standalone families.

FAMILY 1: SYNTHETIC IDENTITY ACCOUNT OPENING

A. Identity

Field	Value
Attack ID	SIA-001
Attack Family	Synthetic Identity Account Opening
Attack Subtypes / Variants	Credit-building / Bust-out accounts; Mule account creation; Synthetic identity farming (incentive abuse / resale); Mass bot-driven account creation
Primary Lifecycle Stage	Account Creation / Onboarding (Stage 2)
Secondary / Cross-stage Stages	Identity / KYC (Stage 1); Payment Initiation (Stage 5); Settlement (Stage 13); Cash-out / Mule (Stage 16)
Target / Actor	Account opening systems, onboarding workflows, KYC verification pipelines, risk engines
Attack Objective	Open one or more operational accounts using synthetic or stolen identities to conduct fraud, launder funds, or resell accounts

B. Fraud Understanding

What is the attack?

The attacker submits a fabricated or partially fabricated identity (synthetic persona) or uses a real victim’s identity with AI-generated supporting evidence to open a fully verified account. The account then serves as a vehicle for downstream fraud: credit bust-outs, money laundering (mules), incentive harvesting, or resale to other criminals via account farms.

Traditional mechanism

Before GenAI, fraudsters manually assembled synthetic identities using breached data (e.g., real SSNs with fake names) and forged documents using Photoshop or physical counterfeiting. Account opening was manual or semi-automated, and each identity required significant effort to cultivate. Account farming was small-scale, reliant on stolen physical documents or insider collusion.

GenAI transformation

GenAI enables industrial-scale generation of hyperrealistic synthetic personas with coherent digital footprints, AI-generated identity documents that pass verification, and automated onboarding flow navigation. Mastercard Q1 2025 data shows synthetic identities surged 300%; Sumsub reports synthetic document fraud rose over 300% YoY in the U.S. Attackers can now produce a convincing synthetic identity in minutes, not weeks.

Why GenAI is load-bearing

If GenAI is removed, the attacker loses the ability to:

- Generate thousands of unique synthetic personas rapidly with consistent attributes
- Create convincing digital footprints (social media, online history) for each identity
- Produce flawless identity documents without specialized design skills
- Automate the cultivation process across multiple identities simultaneously
- Scale the attack from dozens to thousands of accounts

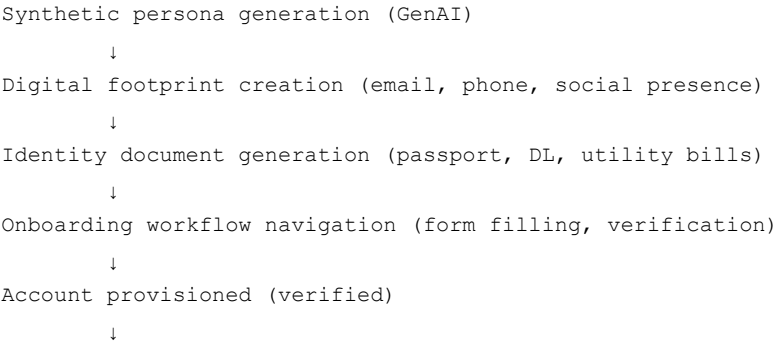
The attack transforms from a niche, labour-intensive operation into an automated industrial campaign.

Prerequisites

- Access to GenAI tools (LLMs, image generation models, document generators)

- Breached PII datasets (real identity fragments) – or purely fabricated data
- Understanding of target institution's onboarding logic
- (For farming) Access to account marketplaces

Attack flow



Variant-specific downstream actions:

• Credit-building → bust-out (30-365 days)

• Mule → receive & layer fraudulent funds

• Farming → resell account on marketplace

• Incentive abuse → harvest bonuses

Downstream payment consequence

The account is treated as legitimate by the institution. This enables:

- Credit card bust-outs (large transactions, maxed credit, default)
- Money laundering via mule accounts (layering, cash-out)
- Fraud-as-a-service (farmed accounts sold to other criminals)
- Incentive harvesting (direct financial loss from bonuses/referrals)

C. Detection Intelligence

Signal Category	Specific Signals	Detection Method
Identity/KYC	Synthetic persona inconsistencies (e.g., mismatched DOB/address history)	ML anomaly detection
	Identity attributes shared across "unrelated" accounts (phone, email domain, address)	Graph analytics / Entity resolution
	Documents with GenAI artifact patterns (pixel-level anomalies, metadata)	ML document forensics
	Inconsistent identity data across application fields	Rule-based + ML
Device/Session	Device fingerprint anomalies / emulator detection	Rule-based + ML
	Multiple applications from same device/IP	Velocity rules
	Virtual camera or automated browser detection	Rule-based + ML
Behavioral	Unnatural form-filling patterns (too fast, no typos, linear)	ML (behavioral biometrics)
	Identical dormancy-to-activity patterns across accounts	ML (temporal clustering)
	Too-rapid credit building timeline	ML (velocity analysis)
Transaction	Transaction amounts inconsistent with declared profile	ML
	Unusual transaction velocity / patterns	ML
	Funds movement to known mule accounts	Graph analytics
	Bust-out transaction sequences (rapid maxing of credit)	ML
Network/Relationship	Shared attributes across "unrelated" accounts (phone, address, email domain)	Graph Neural Network
	Connected to known fraud rings	Consortium intelligence
	Entity resolution graph anomalies	ML (Graph analytics)
Temporal	Accounts with identical dormancy-to-activity patterns	ML

Signal Category	Specific Signals	Detection Method
	Unusual transaction timing (off-hours)	Rule-based + ML
	Too-rapid verification completion	ML

Evasion Considerations

An adaptive attacker could:

- Use unique phone numbers, addresses, and email domains per synthetic identity (AI-generated)
- Create convincing social media histories and online presences for each identity
- Vary dormancy periods randomly (not predictable patterns)
- Use device emulation and randomised fingerprints for each application
- Generate documents with unique GenAI artifact patterns per submission
- Mimic legitimate user behavior (typing speed, mouse movements, navigation patterns)

D. Simulation Intelligence

Field	Value
Simulation type	Algorithmic (primary) – identity and transaction pattern generation; Hybrid for farming/bot-driven variants
Generation parameters	Identity data distributions (name, DOB, address, occupation, income); Document type distribution; Document quality levels; Credit-building sequences; Transaction pattern distributions; Temporal patterns; Network relationship generation; Bot behavior parameters
Synthetic entities required	Synthetic personas (name, DOB, address, occupation, income, email, phone); Synthetic documents (passport, DL, Aadhaar, utility bills); Digital footprint (social media posts, email history); Application form data (filled fields, timing); Transaction histories
Synthetic transactions required	Dormancy period (no transactions); Normal activity (salary deposits, bill payments, small purchases); Credit-building (loan applications, repayments, limit increases); Bust-out (large transactions, maxed credit); Funds movement to mule accounts; Layering transactions
Temporal behavior	Gradual credit-building over 30–365 days; Normal-looking transaction patterns during cultivation; Sudden spike at bust-out; Randomised dormancy durations; Application timing (off-hours weighted for bot variants)
Network relationships	Shared attributes across multiple synthetic identities (farming clusters); Connections to mule accounts; Fraud ring clustering; Marketplace listing patterns (for farming)
Legitimate counterpart	Thin-file customers (new immigrants, students); Legitimate credit-building customers; Users with limited digital history; Customers with legitimate multiple accounts (e.g., joint accounts)
Edge cases	Partially verified identities (NFTF accounts); Identities that pass initial KYC but fail later verification; Varying cultivation durations; Accounts opened but never activated

E. Evidence

Claim	Source	Confidence
Synthetic identities surged 300% in Q1 2025	Mastercard data (ItaliaOggi)	Verified
Synthetic identity document fraud rose 300%+ in U.S. Q1 2025	Sumsb Q1 2025 Identity Fraud Trends	Verified
61% of leaders see synthetic identity as fastest-growing threat	Mastercard 2025 payment fraud prevention report	Verified
Synthetic identities can pass initial checks and build credibility over time	PYMNTS/Mastercard/Trulioo report	Verified
Deloitte projects \$23B losses by 2030	Deloitte Center for Financial Services	Verified
Over 90% of Microsoft account creation attempts in H1 2025 driven by malicious bots	Microsoft Digital Defense Report	Verified
Account farms selling pre-verified accounts across 3,000+ platforms	Resistant AI / Cifas	Verified
360,000+ document templates from 15,000+ issuers circulate on fraud websites	Resistant AI	Verified

F. Product Mapping

Component	Implementation
Generate module	Algorithmic generation of synthetic personas with consistent attributes; Transaction sequence generation for cultivation, bust-out, mule activity, and farming; Bot behavior simulation for mass application; Network relationship generation for synthetic identity clusters; Document generation with varying artifact patterns
Defend module	Entity resolution graph; Identity data cross-reference (CKYCR, DigiLocker); Document forensics; Device fingerprint analysis; Behavioral biometrics; Transaction anomaly detection
Detection model	Graph Neural Network for entity resolution; Anomaly detection for credit-building velocity; Behavioral biometrics for onboarding; CNN/Transformer for document forgery; ML classifier for transaction anomalies
Potential mitigation	Real-time identity verification against trusted sources; Continuous trust scoring (not just event-based); Open finance data sharing for identity validation; Delayed activation for high-risk accounts; Consortium intelligence sharing
Evaluation metric	Precision (avoiding false positives on thin-file customers); Recall (catching synthetic identities before bust-out); AUC; False Positive Rate on legitimate accounts

FAMILY 2: MISREPRESENTATION / DOCUMENT FABRICATION (FIRST-PARTY)

A. Identity

Field	Value
Attack ID	MDF-001
Attack Family	Misrepresentation / Document Fabrication
Attack Subtypes / Variants	Income misrepresentation; Employment fabrication; Address falsification; Occupation fraud; Synthetic financial history
Primary Lifecycle Stage	Account Creation / Onboarding (Stage 2)
Secondary / Cross-stage Stages	Account Limits (Stage 3 – implicit); Payment Initiation (Stage 5); Credit underwriting
Target / Actor	Account opening systems, underwriting engines, credit decisioning, risk scoring
Attack Objective	A legitimate applicant fabricates or inflates financial/occupational attributes using GenAI-generated documents to qualify for better products, higher credit limits, or account access they would not otherwise receive

B. Fraud Understanding

What is the attack?

A real person (first-party) applies for an account or financial product and uses GenAI to generate convincing fabricated supporting documents – paystubs, bank statements, utility bills, employment verification letters – that misrepresent their income, employment, address, or financial history. Unlike Synthetic Identity Fraud, the identity is legitimate; the *attributes* are fraudulent.

Traditional mechanism

Before GenAI, first-party misrepresentation involved manually inflating income on application forms, creating fake paystubs with basic office software (detectable by formatting errors), or providing inconsistent verbal explanations. These were relatively easy for underwriters to detect and difficult to scale.

GenAI transformation

GenAI enables applicants to generate professional-grade fabricated documents with perfect formatting, authentic-looking logos, consistent data across multiple documents, and realistic aging/noise effects – all in seconds. Multimodal LLMs can now generate entire financial histories (bank statements, tax returns) that appear completely legitimate.

Why GenAI is load-bearing

If GenAI is removed, the attacker (first-party) loses the ability to:

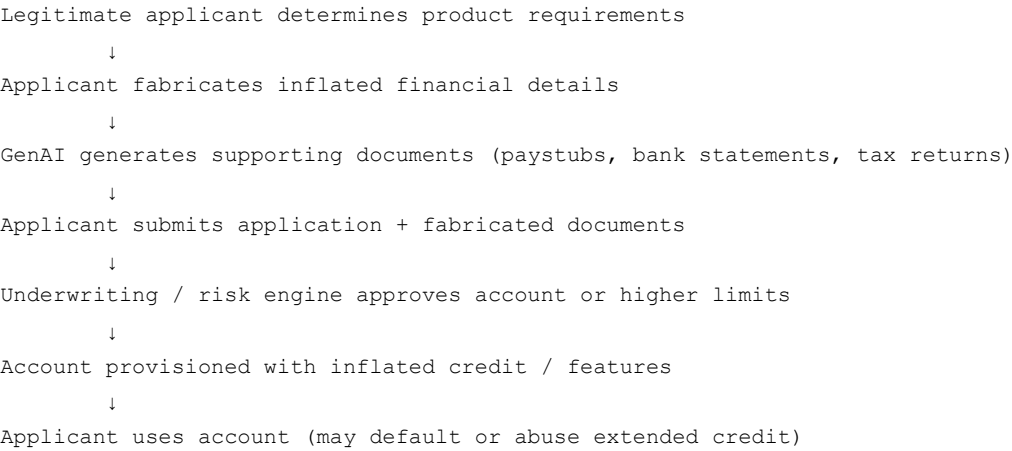
- Generate high-quality fabricated financial documents without specialised graphic design skills
- Create consistent, coherent documentation across multiple document types (paystub + bank statement + tax return)
- Produce documents at scale for multiple applications
- Iteratively improve documents based on rejection feedback

The attack shifts from crude, detectable lies to professionally fabricated financial profiles that pass automated and manual review.

Prerequisites

- Access to GenAI tools (LLMs, image generation, document generation)
- Basic knowledge of the target institution's underwriting requirements
- No PII breach required – the applicant uses their own identity

Attack flow



Downstream payment consequence

The institution extends credit or provides account features based on fabricated financials. Losses occur when:

- The applicant defaults on extended credit (first-party default)
- The applicant abuses higher transaction limits for layering
- The inflated profile enables access to products (e.g., business accounts) used for fraud

C. Detection Intelligence

Signal Category	Specific Signals	Detection Method
Identity/KYC	Inconsistency between declared income and known identity profile (e.g., occupation)	Rule-based + ML
	Document anomalies (font inconsistencies, logo artifacts, metadata)	ML document forensics
	Inconsistency across multiple documents (e.g., paystub vs. bank statement totals)	Rule-based + ML
	OCR data inconsistencies (e.g., math that doesn't add up)	Rule-based
Device/Session	Multiple applications from same device (bonus abuse)	Rule-based
	Unusual submission timing	Rule-based
Behavioral	Unnatural application flow (too fast, no corrections)	ML (behavioral biometrics)
Transaction	Subsequent transaction patterns inconsistent with declared profile	ML
	Rapid credit usage (bust-out following inflated approval)	ML
Network/Relationship	Documents containing same GenAI artifact patterns (fingerprinting)	ML
	Shared fabricated document templates across applications	Document fingerprinting
Temporal	Suspiciously rapid verification / underwriting approval	Rule-based + ML

Evasion Considerations

An adaptive attacker could:

- Add realistic aging/noise effects to fabricated documents
- Generate documents with varied metadata (different creation software, dates)
- Ensure all documents are numerically consistent across types
- Mimic legitimate application behavior (typing speed, corrections)
- Submit during regular business hours to avoid timing flags

D. Simulation Intelligence

Field	Value
Simulation type	Algorithmic
Generation parameters	Income distribution (inflated vs. declared); Document type selection; Document quality levels; Consistency checks; Application timing; Fabrication degree (mild → extreme)
Synthetic entities required	Legitimate applicant personas (name, occupation, real income baseline); Fabricated financial documents (paystubs, bank statements, tax returns, employment letters); Application form data
Synthetic transactions required	Post-account transaction sequences consistent with inflated profile; Potential default / bust-out sequences; Normal legitimate transactions for baseline
Temporal behavior	Document generation timing; Application submission timing; Post-activation transaction ramp-up; Default timing (if applicable)
Network relationships	Document template reuse; Same applicant applying to multiple institutions (fabricated profile across institutions)
Legitimate counterpart	Applicants with legitimate high income; Applicants with legitimate complex financial situations; Applicants with poor-quality but genuine documents
Edge cases	Partial fabrication (some genuine, some fabricated); Self-employed individuals (complex income validation); Applicants with multiple income sources

E. Evidence

Claim	Source	Confidence
GenAI can generate convincing financial documents (paystubs, bank statements)	Industry analysis / Open-source demonstrations	Verified (capability)
1st-party fraud losses are significant in credit products	Industry fraud reporting	Reasonable inference
85% of financial institutions experience fraud in account opening	TransUnion / LexisNexis	Verified
Generative AI models lowered the barrier to document fraud across use cases	FTI Consulting / Fintech security research	Reasonable inference

F. Product Mapping

Component	Implementation
Generate module	Algorithmic generation of fabricated financial documents with varying quality levels; Parameterisation by income, employment type, document type; Consistency generation across document sets
Defend module	Document forensics (font, logo, metadata analysis); Cross-document consistency checks; Income/occupation consistency scoring; External data enrichment (e.g., employer verification)
Detection model	CNN/Transformer for document forgery; Anomaly detection for cross-document inconsistencies; ML classifier for income vs. profile mismatch; Rule-based consistency scoring
Potential mitigation	Third-party income/employment verification; Open finance data sharing; Document fingerprint database; Enhanced underwriting for high-limit accounts
Evaluation metric	True Positive Rate (catching fabricated documents); False Positive Rate (not rejecting genuine documents); Precision/Recall

FAMILY 3: ACCOUNT TAKEOVER VIA ONBOARDING EXPLOITATION

A. Identity

Field	Value
Attack ID	ATO-002

Field	Value
Attack Family	Account Takeover via Onboarding Exploitation
Attack Subtypes / Variants	Deepfake biometric bypass (liveness/activation); AI-generated document recovery fraud; Customer support impersonation; OTP/SIM-swap assisted activation takeover
Primary Lifecycle Stage	Account Creation / Onboarding (Stage 2 – Activation / Recovery)
Secondary / Cross-stage Stages	Authentication (Stage 4); Identity / KYC (Stage 1); Customer Support (Stage 18)
Target / Actor	Account activation flows; Password reset / recovery verification; Biometric verification during activation; Customer support identity verification
Attack Objective	Gain control of a newly created or existing account by impersonating the legitimate user during the onboarding/activation or recovery process using GenAI-generated evidence

B. Fraud Understanding

What is the attack?

The attacker exploits the activation or recovery phase of onboarding – where KYC is re-verified for password resets, device binding, or biometric activation – to take over an account. Using GenAI-generated forged documents, voice clones, or deepfake video, the attacker convinces the institution's verification system (automated or human) that they are the legitimate account holder.

Traditional mechanism

Before GenAI, account takeover via recovery relied on stolen credentials, guessing security questions, SIM swapping, or social engineering of support staff. These required direct access to victim PII and were limited by the need for physical or manual interaction.

GenAI transformation

GenAI enables attackers to generate convincing impersonation evidence on demand:

- Forged identity documents matching the victim's profile (from data breaches)
- Voice clones to pass phone-based verification (voice biometrics)
- Deepfake video to pass video-based recovery verification (V-CIP)

iProov/MITRE ATLAS documented camera injection attacks using face-swap tools and virtual cameras on non-rooted Android devices targeting mobile onboarding. Deepfakes now account for 20% of biometric fraud attempts; injection attacks increased 40% YoY.

Why GenAI is load-bearing

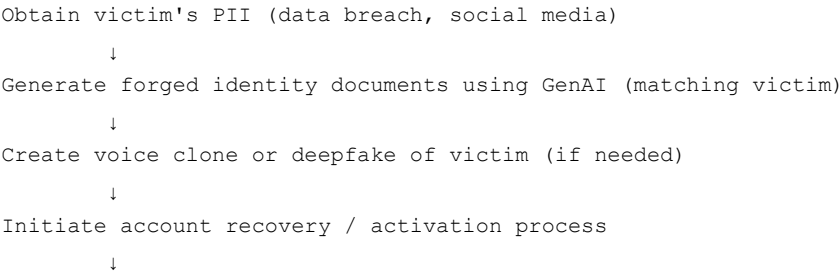
If GenAI is removed, the attacker loses the ability to:

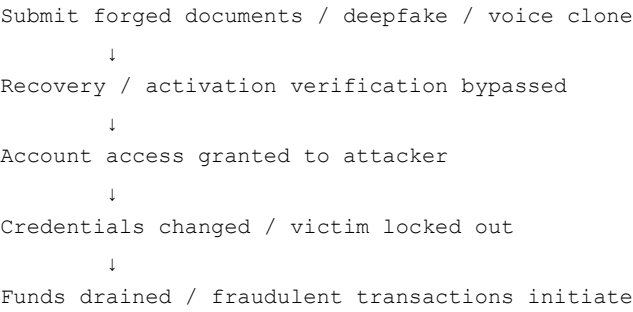
- Generate convincing forged identity documents without physical access to victim's documents
- Create voice clones on demand to pass voice biometrics
- Produce deepfake video for video-based verification
- Perform camera injection attacks (digital sensor bypass) vs. physical presentation attacks

Prerequisites

- Victim's PII (from data breaches, social media, phishing)
- Access to GenAI tools (document generation, voice cloning, deepfake video)
- Knowledge of the target institution's recovery/activation verification process
- Camera injection capability (for deepfake activation)

Attack flow





Downstream payment consequence

Account takeover leads to:

- Full control of victim’s financial account
- Funds drained (transfers to mule accounts)
- New beneficiaries added for ongoing fraud
- Fraudulent transactions authorized
- Additional synthetic accounts opened using the compromised account as a "verified anchor"
- Long-term reputation damage to the victim

C. Detection Intelligence

Signal Category	Specific Signals	Detection Method
Identity/KYC	Document anomalies in recovery submissions	ML document forensics
	Biometric match anomalies (e.g., liveness failure)	ML
	Inconsistent identity data (recovery details vs. original KYC)	Rule-based + ML
	Deepfake artifact patterns in video	ML (CNN/Transformer)
Device/Session	Unusual device for recovery/activation request	Rule-based
	Unusual location/IP	Rule-based + ML
	Multiple recovery attempts in short time	Velocity rules
	Virtual camera detection	Hardware-level / ML
	Device emulation detection	Rule-based
Behavioral	Unnatural interaction patterns (too fast, linear)	ML (behavioral biometrics)
	Signs of automated navigation	ML
	Lack of micro-expressions in video	ML
	Audio-visual desynchronization	ML
Transaction	Unauthorized transactions post-recovery	ML
	New beneficiary additions	Rule-based + ML
	Funds movement to mule accounts	Graph analytics
Network/Relationship	Recovery attempts connected to known fraud patterns	ML
	Same deepfake artifact patterns across recovery attempts	Document/Video fingerprinting
Temporal	Recovery at unusual times (off-hours)	Rule-based
	Rapid completion of recovery flow	ML
	Time between recovery and first fraudulent transaction	Rule-based

Evasion Considerations

An adaptive attacker could:

- Use high-quality forged documents with realistic noise/aging

- Use realistic voice clones with natural speech patterns
- Add realistic micro-expressions and natural head movements to deepfakes
- Mimic victim's normal behavior post-recovery
- Use victim's own device (if compromised) to avoid device flags
- Rotate virtual camera configurations to evade detection
- Use unique deepfake patterns for each attempt

D. Simulation Intelligence

Field	Value
Simulation type	Agentic (primary) – deepfake/voice generation and verification interactions; Hybrid with algorithmic document generation
Generation parameters	Deepfake quality levels; Voice clone quality; Document forgery quality; Injection method variants (camera, audio, file upload); Recovery channel selection; Interaction sequences
Synthetic entities required	Victim personas (legitimate account holders with transaction histories); Forged identity documents (victim's details); Deepfake video streams; Voice clone audio; Recovery interaction sequences; Device/session data showing injection attempt
Synthetic transactions required	Post-recovery unauthorized transactions; New beneficiary additions; Funds movement to mule accounts; Rapid transaction sequences
Temporal behavior	Recovery timing (off-hours weighted); Time between recovery and fraudulent activity; Transaction velocity patterns; Verification duration (abnormal if too fast)
Network relationships	Recovery attempts connected to known fraud patterns; Same deepfake artifact signature across attempts; Mule network connections
Legitimate counterpart	Legitimate account recovery (lost password, forgotten security questions); Legitimate activation (new device binding); Users with poor camera/lighting quality; Users with disabilities affecting liveness response
Edge cases	Partial recovery attempts (some checks pass, some fail); Multiple recovery channels; Different recovery verification requirements; Injection attempts that are partially successful (some checks pass, some fail)

E. Evidence

Claim	Source	Confidence
Deepfakes account for 20% of biometric fraud attempts	Entrust 2026 Identity Fraud Report	Verified
Deepfake selfies increased 58% in 2025	Entrust 2026 Identity Fraud Report	Verified
Injection attacks increased 40% YoY	Entrust 2026 Identity Fraud Report	Verified
Crypto targeted in 60% of deepfake cases	Entrust 2026 Identity Fraud Report	Verified
MITRE ATLAS published iProov deepfake KYC injection case study	MITRE ATLAS / iProov	Verified
Attack uses Faceswap + OBS + Virtual Camera on non-rooted Android	iProov / MITRE ATLAS	Verified
FATF named liveness check as attack surface	FATF Horizon Scan (22 Dec 2025)	Verified
RBI August 2025 amendment mandated active deepfake/spoof detection in V-CIP	RBI KYC Master Direction amendment	Verified
Account takeover incidents rose 21% YoY in H1 2025	Entrust / Industry reports	Verified

F. Product Mapping

Component	Implementation
Generate module	Agentic generation of deepfake video/audio content; Algorithmic generation of forged documents; Hybrid generation of verification flow interactions; Device/session data generation
Defend module	AI-driven deepfake detection; Hardware-level sensor validation; Behavioral biometrics; Voice biometrics; Document forensics; Multi-factor verification for recovery
Detection model	Deepfake artifact detection (CNN/Transformer); Liveness detection with multi-factor verification; Behavioral biometrics for facial movements; Voice biometric matching; Anomaly detection for recovery attempts

Component	Implementation
Potential mitigation	Multi-layer verification (hardware-based + hybrid liveness); AI-driven content validation tools; Human expertise for contextual anomalies; Enhanced verification for recovery (additional factors); Delayed recovery processing for high-risk cases; Customer notification of recovery attempts
Evaluation metric	False Acceptance Rate (FAR); False Rejection Rate (FRR); Speed (real-time verification required); Detection Rate

ATTACK × OBSERVABLE SIGNAL MATRIX

Attack Family	Identity/KYC	Device/Session	Behavioral	Transaction	Network/Relationship	Temporal
Synthetic Identity Account Opening	Synthetic persona inconsistencies; Shared attributes across accounts; GenAI document artifacts; Inconsistent identity data	Emulator detection; Multiple applications from same device; Virtual camera/bot detection	Unnatural form-filling; Identical dormancy patterns; Too-rapid credit building	Amounts inconsistent with profile; Unusual velocity; Funds to mule accounts; Bust-out sequences	Shared attributes across "unrelated" accounts; Connected to fraud rings; Entity resolution anomalies	Identical dormancy patterns; Unusual timing; Too-rapid verification
Misrepresentation / Document Fabrication	Income vs. profile mismatch; Document artifacts; Cross-document inconsistencies; OCR inconsistencies	Multiple applications (bonus abuse); Unusual submission timing	Unnatural application flow; No corrections	Post-account transactions inconsistent with declared profile; Rapid credit usage	Same template artifacts across docs	Suspiciously rapid underwriting
Account Takeover via Onboarding Exploitation	Document anomalies in recovery; Biometric match anomalies; Deepfake artifacts; Inconsistent identity data	Unusual device/IP; Multiple recovery attempts; Virtual camera detection; Device emulation	Unnatural interactions; Lack of micro-expressions; Audio-visual desync; Automated navigation	Unauthorized transactions post-recovery; New beneficiary additions; Funds to mules	Recovery attempts connected to fraud patterns; Same artifact signatures	Off-hours recovery; Rapid verification; Time-to-fraud short

ATTACK × SIMULATION PARAMETER MATRIX

Attack Family	Identity Attributes	Document Attributes	Device/Session Attributes	Temporal Parameters	Transaction Sequences	Network Relationships	Behavioral Parameters
Synthetic Identity Account Opening	Name, DOB, address, occupation, income, email, phone (distributions)	Type (passport/DL/bill); Quality (pristine/aged/scanned); Artifact patterns; Metadata	Fingerprint, IP, geolocation; Emulator flags; Bot pacing	Cultivation duration (30–365 days); Dormancy; Application timing; Transaction timing	Cultivation → normal → bust-out / mule / layering; Credit-building sequences	Shared attributes (clusters); Fraud ring connections; Marketplace listings	Form-filling speed; Typos; Navigation patterns
Misrepresentation / Document Fabrication	Real identity profile; Declared vs. actual income/occupation	Fabricated financial documents (paystub, bank statement, tax return); Quality levels; Consistency	Submission timing; Application timing	Document generation timing; Application timing; Post-account ramp-up; Default timing	Normal transactions (baseline) vs. inflated credit usage	Document template reuse; Same applicant across institutions	Application flow naturalness
Account Takeover via Onboarding Exploitation	Victim PII (DOB, address, SSN); Account details	Forged recovery documents; Deepfake video; Voice clone audio; Quality levels	Virtual camera/audio; Device spoofing; IP/location spoofing	Recovery timing (off-hours); Verification duration; Time to fraud	Post-recovery fraud sequence; Beneficiary additions; Funds movement	Recovery attempts connected to fraud patterns; Same deepfake signatures	Natural speech/movement; Micro-expressions; Interaction latency

FINAL RECOMMENDATIONS

Candidate (from Pass C / Audit)	Action	Rationale
Synthetic Identity Account Opening	KEEP (Distinct Family)	GenAI is load-bearing for industrial-scale identity generation; primary account creation attack vector
Misrepresentation / Document Fabrication	KEEP (Distinct Family)	First-party fraud is materially different from identity fraud; GenAI transforms document fabrication from detectable to undetectable
Account Takeover via Onboarding Exploitation	KEEP (Distinct Family)	Cross-stage with Authentication but critical to the onboarding activation/recovery surface; deepfake injection is a qualitatively new vector
Account Farming / Factory	MERGE → Variant of Family 1	Farming is the commercialization of Synthetic Identity Account Opening; not a distinct attack mechanism
Synthetic Mule Account Opening	MERGE → Variant of Family 1	Mule creation is a purpose-driven variant of Synthetic Identity Account Opening; mechanism is identical
Identity Theft-Based Account Opening	MERGE → Variant of Family 1	Uses a real victim's identity with AI-generated documents; mechanism overlaps significantly with Synthetic Identity; distinguish by variant, not family
Mass Synthetic Account Creation (Incentive Abuse)	MERGE → Variant of Family 1	Bot-driven incentive farming is a farming subtype; not a separate family
Application Fraud (3rd-party)	MERGE → Variant of Family 1	Covered under Identity Theft-Based variant
Application Fraud (1st-party)	MERGE → Family 2	This is Misrepresentation / Document Fabrication — keep as distinct family
Duplicate Account Creation	MERGE → Variant of Family 1	Incentive farming subtype
Bot-Driven Account Creation	MERGE → Variant of Family 1	Mechanism (bots) rather than attack family; covered by Farming variant
Account Takeover (General)	DROP from Account Creation taxonomy	General ATO belongs to Authentication; only the Onboarding/Activation surface version (Family 3) is relevant here
Insider-Assisted Account Fraud	DROP	GenAI is not load-bearing; removed in Pass C
GenAI-Optimized Risk Score Evasion	NEEDS MORE RESEARCH	Theoretically plausible but no documented real-world cases; track as an <i>advanced simulation capability</i> (Generate phase) rather than a named attack family. Do not include in taxonomy v1.
Synthetic Peer/Employer Verification	NEEDS MORE RESEARCH	No documented cases; treat as a simulation variant within Family 1 (Synthetic Identity), not a separate family.

FINAL TAXONOMY SUMMARY

#	Attack Family	GenAI Load-Bearing	Simulation Type	Primary Stage	Distinctness
1	Synthetic Identity Account Opening	PASS — industrial-scale generation	Algorithmic (Hybrid for bot/farming)	Account Creation	Distinct family
2	Misrepresentation / Document Fabrication	PARTIAL — materially improves, but 1st-party fraud existed	Algorithmic	Account Creation	Distinct family
3	Account Takeover via Onboarding Exploitation	PASS — deepfake injection is qualitatively new	Agentic (Hybrid for documents)	Account Creation ∩ Authentication	Distinct family

Total: 3 distinct families, with multiple variants nested under Family 1. This mirrors the structure of the KYC taxonomy (5 families) while maintaining the same depth of analysis. The grouping balances breadth of coverage with conceptual clarity — every attack surface from the coverage audit is either captured in these 3 families or explicitly merged/dropped with rationale.